```
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % nano test_model.py
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % python test_model.py
/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2
.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
🧠 Loading your custom fine-tuned model from ./m5_adapters...
Fetching 8 files: 100%|████████████████████████████████████████████████████████████| 8/8 [00:00<00:00, 38612.70it/s]

🚀 Running Inference on M5 GPU...
Traceback (most recent call last):
  File "/Users/lisara/Desktop/desk-triage-local/test_model.py", line 29, in <module>
    output = generate(model, tokenizer, prompt=prompt, temperature=0.0, max_tokens=200)
  File "/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/mlx_lm/generate.py", line 761, in generate
    for response in stream_generate(model, tokenizer, prompt, **kwargs):
  File "/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/mlx_lm/generate.py", line 685, in stream_generate
    token_generator = generate_step(prompt, model, **kwargs)
TypeError: generate_step() got an unexpected keyword argument 'temperature'
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % nano test_model.py
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % python test_model.py
/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2
.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
🧠 Loading your custom fine-tuned model from ./m5_adapters...
Fetching 8 files: 100%|████████████████████████████████████████████████████████████| 8/8 [00:00<00:00, 96420.78it/s]

🚀 Running Inference on M5 GPU...

🤖 Model Output:
========================================
{"":{"": "Unknown", "priority": "Medium", "llm_draft_reply": ""}
========================================
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % nano prepare_data.py
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % python prepare_data.py
/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2
.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
📡 Downloading dataset from Hugging Face Hub...
✂️Splitting data...
✅ Done! Raw pre-formatted text saved.
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % python -m mlx_lm lora \
  --model mlx-community/Meta-Llama-3.1-8B-Instruct-4bit \
  --data . \
  --train \
  --learning-rate 2e-4 \
  --batch-size 2 \
  --iters 400 \
  --save-every 100 \
  --grad-checkpoint \
  --adapter-path ./m5_adapters_8b
/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2
.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
Loading pretrained model
special_tokens_map.json: 100%|█████████████████████████████████████████████████████| 296/296 [00:00<00:00, 145kB/s]
tokenizer_config.json: 55.4kB [00:00, 18.3MB/s]                                         | 0.00/4.52G [00:00<?, ?B/s]
model.safetensors.index.json: 52.4kB [00:00, 40.6MB/s]                                   | 0.00/296 [00:00<?, ?B/s]
config.json: 1.02kB [00:00, 1.39MB/s]?B/s]
tokenizer.json: 9.08MB [00:00, 10.9MB/s]                                               | 1/6 [00:01<00:07,  1.54s/it]
model.safetensors: 100%|████████████████████████████████████████████████████| 4.52G/4.52G [35:24<00:00, 2.13MB/s]
Fetching 6 files: 100%|█████████████████████████████████████████████████████████████| 6/6 [35:25<00:00, 354.25s/it]
Loading datasets
Training
Trainable parameters: 0.131% (10.486M/8030.261M)
Starting training..., iters: 400
Calculating loss...: 100%|█████████████████████████████████████████████████████████| 25/25 [00:45<00:00,  1.84s/it]
Iter 1: Val loss 3.279, Val took 45.956s
Iter 10: Train loss 3.100, Learning Rate 2.000e-04, It/sec 0.225, Tokens/sec 90.122, Trained Tokens 4005, Peak mem 5.821 GB
Iter 20: Train loss 18.267, Learning Rate 2.000e-04, It/sec 0.291, Tokens/sec 94.840, Trained Tokens 7261, Peak mem 5.821 GB
Iter 30: Train loss 9.751, Learning Rate 2.000e-04, It/sec 0.251, Tokens/sec 89.031, Trained Tokens 10802, Peak mem 6.038 GB
Iter 40: Train loss 8.534, Learning Rate 2.000e-04, It/sec 0.256, Tokens/sec 90.126, Trained Tokens 14323, Peak mem 6.038 GB
Iter 50: Train loss 8.227, Learning Rate 2.000e-04, It/sec 0.235, Tokens/sec 91.377, Trained Tokens 18212, Peak mem 6.038 GB
Iter 60: Train loss 7.808, Learning Rate 2.000e-04, It/sec 0.263, Tokens/sec 96.539, Trained Tokens 21876, Peak mem 6.038 GB
Iter 70: Train loss 7.833, Learning Rate 2.000e-04, It/sec 0.231, Tokens/sec 85.747, Trained Tokens 25591, Peak mem 6.038 GB
Iter 80: Train loss 7.574, Learning Rate 2.000e-04, It/sec 0.275, Tokens/sec 92.395, Trained Tokens 28955, Peak mem 6.038 GB
Iter 90: Train loss 7.619, Learning Rate 2.000e-04, It/sec 0.267, Tokens/sec 91.236, Trained Tokens 32374, Peak mem 6.038 GB
Iter 100: Train loss nan, Learning Rate 2.000e-04, It/sec 0.234, Tokens/sec 92.167, Trained Tokens 36317, Peak mem 6.038 GB
Iter 100: Saved adapter weights to m5_adapters_8b/adapters.safetensors and m5_adapters_8b/0000100_adapters.safetensors.
Iter 110: Train loss nan, Learning Rate 2.000e-04, It/sec 0.266, Tokens/sec 93.474, Trained Tokens 39837, Peak mem 6.038 GB
Iter 120: Train loss nan, Learning Rate 2.000e-04, It/sec 0.255, Tokens/sec 91.980, Trained Tokens 43443, Peak mem 6.038 GB
Iter 130: Train loss nan, Learning Rate 2.000e-04, It/sec 0.236, Tokens/sec 89.887, Trained Tokens 47248, Peak mem 6.038 GB
^CTraceback (most recent call last):
  File "/Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/runpy.py", line 197, in _run_module_as_main
    return _run_code(code, main_globals, None,
  File "/Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/runpy.py", line 87, in _run_code
    exec(code, run_globals)
  File "/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/mlx_lm/__main__.py", line 30, in <module>
    submodule.main()
  File "/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/mlx_lm/lora.py", line 362, in main
    run(types.SimpleNamespace(**args))
  File "/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/mlx_lm/lora.py", line 334, in run
    train_model(args, model, train_set, valid_set, training_callback)
  File "/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/mlx_lm/lora.py", line 288, in train_model
    train(
  File "/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/mlx_lm/tuner/trainer.py", line 312, in train
    mx.eval(state, losses, n_tokens, grad_accum)
KeyboardInterrupt

(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % python -m mlx_lm lora \
  --model mlx-community/Meta-Llama-3.1-8B-Instruct-4bit \
  --data . \
  --train \
  --learning-rate 5e-5 \
  --batch-size 2 \
  --iters 400 \
  --save-every 100 \
  --grad-checkpoint \
  --adapter-path ./m5_adapters_8b
/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2
.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
Loading pretrained model
Fetching 6 files: 100%|██████████████████████████████████████████████████████████████████████████████████████████████████████| 6/6 [00:00<00:00, 10600.60it/s]
Loading datasets
Training
Trainable parameters: 0.131% (10.486M/8030.261M)
Starting training..., iters: 400
```

````
Calculating loss...: 100%|████████████████████████████████████████| 25/25 [00:45<00:00,  1.83s/it]
Iter 1: Val loss 3.279, Val took 45.703s
Iter 10: Train loss 1.847, Learning Rate 5.000e-05, It/sec 0.230, Tokens/sec 92.165, Trained Tokens 4005, Peak mem 5.821 GB
Iter 20: Train loss 1.103, Learning Rate 5.000e-05, It/sec 0.293, Tokens/sec 95.348, Trained Tokens 7261, Peak mem 5.821 GB
Iter 30: Train loss 1.124, Learning Rate 5.000e-05, It/sec 0.252, Tokens/sec 89.128, Trained Tokens 10802, Peak mem 6.038 GB
Iter 40: Train loss 1.085, Learning Rate 5.000e-05, It/sec 0.260, Tokens/sec 91.423, Trained Tokens 14323, Peak mem 6.038 GB
Iter 50: Train loss 1.103, Learning Rate 5.000e-05, It/sec 0.237, Tokens/sec 92.005, Trained Tokens 18212, Peak mem 6.038 GB
Iter 60: Train loss 1.026, Learning Rate 5.000e-05, It/sec 0.262, Tokens/sec 96.066, Trained Tokens 21876, Peak mem 6.038 GB
Iter 70: Train loss 1.087, Learning Rate 5.000e-05, It/sec 0.230, Tokens/sec 85.467, Trained Tokens 25591, Peak mem 6.038 GB
Iter 80: Train loss 0.960, Learning Rate 5.000e-05, It/sec 0.270, Tokens/sec 90.861, Trained Tokens 28955, Peak mem 6.038 GB
Iter 90: Train loss 1.013, Learning Rate 5.000e-05, It/sec 0.267, Tokens/sec 91.251, Trained Tokens 32374, Peak mem 6.038 GB
Iter 100: Train loss 1.067, Learning Rate 5.000e-05, It/sec 0.239, Tokens/sec 94.174, Trained Tokens 36317, Peak mem 6.038 GB
Iter 100: Saved adapter weights to m5_adapters_8b/adapters.safetensors and m5_adapters_8b/0000100_adapters.safetensors.
Iter 110: Train loss 1.035, Learning Rate 5.000e-05, It/sec 0.270, Tokens/sec 95.077, Trained Tokens 39837, Peak mem 6.038 GB
Iter 120: Train loss 0.987, Learning Rate 5.000e-05, It/sec 0.256, Tokens/sec 92.214, Trained Tokens 43443, Peak mem 6.038 GB
Iter 130: Train loss 1.045, Learning Rate 5.000e-05, It/sec 0.241, Tokens/sec 91.807, Trained Tokens 47248, Peak mem 6.038 GB
Iter 140: Train loss 1.017, Learning Rate 5.000e-05, It/sec 0.220, Tokens/sec 91.687, Trained Tokens 51421, Peak mem 6.038 GB
Iter 150: Train loss 0.985, Learning Rate 5.000e-05, It/sec 0.275, Tokens/sec 96.731, Trained Tokens 54939, Peak mem 6.038 GB
Iter 160: Train loss 1.069, Learning Rate 5.000e-05, It/sec 0.232, Tokens/sec 85.631, Trained Tokens 58624, Peak mem 6.038 GB
Iter 170: Train loss 1.032, Learning Rate 5.000e-05, It/sec 0.236, Tokens/sec 93.890, Trained Tokens 62596, Peak mem 6.038 GB
Iter 180: Train loss 1.042, Learning Rate 5.000e-05, It/sec 0.245, Tokens/sec 89.361, Trained Tokens 66240, Peak mem 6.038 GB
Iter 190: Train loss 1.046, Learning Rate 5.000e-05, It/sec 0.220, Tokens/sec 92.146, Trained Tokens 70427, Peak mem 6.080 GB
Calculating loss...: 100%|████████████████████████████████████████| 25/25 [00:47<00:00,  1.91s/it]
Iter 200: Val loss 1.006, Val took 47.742s
Iter 200: Train loss 0.963, Learning Rate 5.000e-05, It/sec 0.260, Tokens/sec 89.915, Trained Tokens 73882, Peak mem 6.080 GB
Iter 200: Saved adapter weights to m5_adapters_8b/adapters.safetensors and m5_adapters_8b/0000200_adapters.safetensors.
Iter 210: Train loss 0.994, Learning Rate 5.000e-05, It/sec 0.224, Tokens/sec 84.291, Trained Tokens 77643, Peak mem 6.080 GB
Iter 220: Train loss 0.904, Learning Rate 5.000e-05, It/sec 0.251, Tokens/sec 88.852, Trained Tokens 81181, Peak mem 6.080 GB
Iter 230: Train loss 1.052, Learning Rate 5.000e-05, It/sec 0.242, Tokens/sec 91.564, Trained Tokens 84971, Peak mem 6.080 GB
Iter 240: Train loss 1.049, Learning Rate 5.000e-05, It/sec 0.248, Tokens/sec 92.722, Trained Tokens 88713, Peak mem 6.080 GB
Iter 250: Train loss 0.937, Learning Rate 5.000e-05, It/sec 0.267, Tokens/sec 94.362, Trained Tokens 92252, Peak mem 6.080 GB
Iter 260: Train loss 0.976, Learning Rate 5.000e-05, It/sec 0.255, Tokens/sec 90.582, Trained Tokens 95807, Peak mem 6.080 GB
Iter 270: Train loss 0.934, Learning Rate 5.000e-05, It/sec 0.263, Tokens/sec 96.157, Trained Tokens 99467, Peak mem 6.080 GB
Iter 280: Train loss 1.030, Learning Rate 5.000e-05, It/sec 0.236, Tokens/sec 86.149, Trained Tokens 103119, Peak mem 6.080 GB
Iter 290: Train loss 0.915, Learning Rate 5.000e-05, It/sec 0.250, Tokens/sec 88.532, Trained Tokens 106655, Peak mem 6.080 GB
Iter 300: Train loss 0.956, Learning Rate 5.000e-05, It/sec 0.292, Tokens/sec 96.884, Trained Tokens 109968, Peak mem 6.080 GB
Iter 300: Saved adapter weights to m5_adapters_8b/adapters.safetensors and m5_adapters_8b/0000300_adapters.safetensors.
Iter 310: Train loss 1.001, Learning Rate 5.000e-05, It/sec 0.251, Tokens/sec 90.621, Trained Tokens 113572, Peak mem 6.080 GB
Iter 320: Train loss 1.009, Learning Rate 5.000e-05, It/sec 0.244, Tokens/sec 92.374, Trained Tokens 117357, Peak mem 6.080 GB
Iter 330: Train loss 0.943, Learning Rate 5.000e-05, It/sec 0.251, Tokens/sec 87.526, Trained Tokens 120846, Peak mem 6.080 GB
Iter 340: Train loss 0.975, Learning Rate 5.000e-05, It/sec 0.243, Tokens/sec 90.799, Trained Tokens 124586, Peak mem 6.080 GB
Iter 350: Train loss 1.073, Learning Rate 5.000e-05, It/sec 0.214, Tokens/sec 91.470, Trained Tokens 128857, Peak mem 6.172 GB
Iter 360: Train loss 0.913, Learning Rate 5.000e-05, It/sec 0.241, Tokens/sec 87.399, Trained Tokens 132483, Peak mem 6.172 GB
Iter 370: Train loss 0.977, Learning Rate 5.000e-05, It/sec 0.247, Tokens/sec 91.292, Trained Tokens 136179, Peak mem 6.172 GB
Iter 380: Train loss 0.870, Learning Rate 5.000e-05, It/sec 0.271, Tokens/sec 93.353, Trained Tokens 139626, Peak mem 6.172 GB
Iter 390: Train loss 0.943, Learning Rate 5.000e-05, It/sec 0.297, Tokens/sec 95.541, Trained Tokens 142846, Peak mem 6.172 GB
Calculating loss...: 100%|████████████████████████████████████████| 25/25 [00:46<00:00,  1.85s/it]
Iter 400: Val loss 0.960, Val took 46.203s
Iter 400: Train loss 0.973, Learning Rate 5.000e-05, It/sec 0.257, Tokens/sec 93.816, Trained Tokens 146499, Peak mem 6.172 GB
Iter 400: Saved adapter weights to m5_adapters_8b/adapters.safetensors and m5_adapters_8b/0000400_adapters.safetensors.
Saved final weights to m5_adapters_8b/adapters.safetensors.
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % python test_baseline_8b.py
/Users/lisara/Desktop/desk-triage-local/.venv/bin/python: can't open file '/Users/lisara/Desktop/desk-triage-local/test_baseline_8b.py': [Errno 2] No such file or directory
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % nano test_baseline_8b.py
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % python test_baseline_8b.py
/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2
.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
📡 Loading the RAW, UN-TUNED Llama-3.1-8B model from cache...
Fetching 6 files: 100%|████████████████████████████████████████| 6/6 [00:00<00:00, 68947.46it/s]

🚀 Running Baseline Inference on M5 GPU...

📥 === [RAW 8B BASELINE MODEL OUTPUT] ===
```json
{
  "department": "IT",
  "priority": "High",
  "llm_draft_reply": "I've escalated your issue to our IT department. They'll investigate the cause of the HR portal crash and restore access to your employee dashboard. In the meantime, please contact payroll to
confirm your direct deposit details for this month's payroll cycle. We'll also review your paternity leave tracking log to ensure it was saved correctly."
}
```
==========================================

(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % nano test_model.py
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local % python test_model.py
/Users/lisara/Desktop/desk-triage-local/.venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2
.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
🧠 Loading your custom fine-tuned 8B model and adapters...
Fetching 6 files: 100%|████████████████████████████████████████| 6/6 [00:00<00:00, 9049.20it/s]

🚀 Running Fine-Tuned Inference on M5 GPU...

🤖 [FINE-TUNED MODEL OUTPUT]:
========================================
{"department": "IT_Support", "priority": "Critical", "llm_draft_reply": "Dear [Employee's Name],\n\nThank you for your message. We apologize for the inconvenience you are experiencing with the HR portal. Our team
will investigate the issue with the webpage crashing and work on restoring your access to the employee dashboard. Please provide us with any error messages you may have encountered, and we will ensure your direct
deposit details are accurately reflected in the system. \n\nBest regards,\nIT Support Team"}
========================================
(.venv) lisara@Lisaras-MacBook-Pro desk-triage-local %
````